

# Final Project: Preparing Your Mind for Generative AI

Estimated time: 30 minutes

Welcome to the final project!

## Learning objectives

At the end of this project, you will be able to:

- Identify the main concerns of using generative AI models, such as data bias, copyright infringement, and data privacy.
- List measures that organizations can take to implement generative AI responsibly.
- Explore strategies to recognize and address the challenges posed by generative AI.

## Introduction

In this course, we've talked about the widespread impact of generative AI on the economy, the workforce, the environment, and our mental health. While foundation models' capabilities are astounding and relatively easy to learn, users must balance the concerns and challenges that emerge with generative AI. Therefore, it is necessary to prepare your mind to recognize and address the challenges posed by generative AI.

In this final project, you will do exactly that. You will be introduced to certain situations and asked to help determine what next steps should be taken to address emerging issues. Keep in mind that this is not a test but an engaging activity to help you develop a responsive mindset toward generative AI. Just go step-by-step and think about how to respond to each situation.

### Situation 1: What the chatbot says

You are using a generative AI chatbot (for example, ChatGPT, Gemini, Bing, or Jasper Chat) to help you write social media captions for your Instagram page. You like the results so much that you ask the chatbot to write an article on *Camaraderie in the Workplace*. You change a few words around and then post the article you wrote. You never realized using a chatbot could be so easy. A few days later, you get a lot of comments accusing you of being culturally insensitive in your article. You respond by saying that you got this information from a chatbot.

#### Think about this!

1. What is the main concern here? Data bias, copyright infringement, data privacy violation, or human error?
2. What could you have done differently to ensure that the article you posted was more gender-neutral?

### Situation 2: Who said it first

You ask your friend for advice. She advises you to use information from chatbots cautiously. Always read through and rewrite the information to make it your own. While the law on who owns AI-produced data and images is not fully developed, it is being recognized that copy-pasting AI-generated content verbatim is an infringement of copyright.

Plus, many tools are used to check for AI plagiarism, as it is considered unethical to reuse AI-generated data and claim it as your own. Human error is definitely at play here. A best practice is to use generative AI tools for research, ideas, and brainstorming, but not to let it write for you. The final work you create should be your own.

#### Think about this!

1. Do you agree or disagree with this advice?
2. Can you think of other instances where you have used AI-generated content and had a change in perspective?
3. What were the lessons learned?

### Situation 3: Waiting for my turn to come

You and your partner have applied for a mortgage through the bank's online system. You have been rejected for the second time despite having a great credit score and well-paying jobs. You complain about this to your colleague over lunch, and to your surprise, you learn that he and his partner have been instantly approved for a mortgage from the same bank in less than a week.

You call your banker frustrated, asking if the AI software that the bank uses may have been denying mortgages to people of color, even if they have the same financial capacity as a white person. Your banker does not confirm nor deny your query but encourages you to re-apply, hoping the system comes to its senses soon.

#### Think about this!

1. What is your first reaction when you hear this?
2. Is this a concern regarding data bias, copyright infringement, data privacy violation, or human error?
3. What action can you take to be heard and noticed in such a situation?

### Situation 4: Can we retrain an algorithm

Dealing with prejudice is never easy, whether it stems from humans or machines. Unfortunately, the data that foundation models are trained on carries a lot of embedded human biases, leading to algorithmic bias in generative AI systems. Let's look at it this way: the algorithm can only repeat what it has been taught.

As a consumer, you can immediately give feedback to the AI system, questioning possible bias in the screening process. You can ask your bank or whichever institution you are working with to disclose the data sources they are using to train their AI system and ask for full transparency on the AI algorithm's thought process. That is, why did the algorithm reject the same application twice? By giving feedback to the system, you will automatically make it query its findings and train it to think along bias-free criteria.

#### Think about this!

1. How can organizations like the bank curb algorithmic bias in their AI models?

2. Can you list five measures that they can take?

### **Situation 5: Ethics over legality**

The bank's Board of Directors meets and decides to use two immediate measures to query their data sets and locate inherent bias:

- Engage prompt engineers to vet and remove stereotypes and other intentional and unintentional biases from their data
- Seek feedback from users to improve the accuracy and quality of the data.

They are debating whether they should be fully transparent about their data sources and disclose this information to all testers and users. Or whether they should offer explanations to users on how certain decisions and recommendations were made by the algorithm.

#### **Think about this!**

1. How can organizations like the bank implement generative AI more responsibly?
2. What about the law in your country: Is AI regulation pressuring organizations to be more answerable, accountable, transparent, and responsive?

### **Conclusion**

The adoption, usage, and benefits of generative AI depend on the technology being reviewable, trustworthy, and contextualized. Therefore, generative AI models need to be explainable, accurate, and accountable.

For example, users of ChatGPT expect OpenAI to reveal the complex algorithm that ChatGPT uses to respond to queries and prompts. While OpenAI is not legally bound to reveal this information, it may help greatly if AI developers are transparent about their data sources. This would clear up the grey areas of copyright infringement, data privacy violation, and algorithmic bias and make users of AI-generated content more accountable and answerable in how they present AI-generated information.

© IBM Corporation. All rights reserved.



# **Skills Network**